

Predictive Modeling and Analysis of Health Insurance Claims Using Machine Learning

Konjarla Sripriya

*Dept. of Computer Science Engineering,
Amrita School of Computing, Bengaluru
Amrita Vishwa Vidyapeetham, India
bl.en.u4cse22121@bl.students.amrita.edu*

Likhitha Sri Kandukuri

*Dept. of Computer Science Engineering,
Amrita School of Computing, Bengaluru
Amrita Vishwa Vidyapeetham, India
bl.en.u4cse22125@bl.students.amrita.edu*

Kammari Vidyasri

*Dept. of Computer Science Engineering,
Amrita School of Computing, Bengaluru
Amrita Vishwa Vidyapeetham, India
bl.en.u4cse22120@bl.students.amrita.edu*

Dr. AmulyaShree S

*Dept. of Computer Science Engineering,
Amrita School of Computing, Bengaluru
Amrita Vishwa Vidyapeetham, India
s_amulyashree@blr.amrita.edu*

Dr. B. Uma Maheswari

*Dept. of Computer Science Engineering,
Amrita School of Computing, Bengaluru
Amrita Vishwa Vidyapeetham, India
b_uma@blr.amrita.edu*

Abstract—Health insurance plays a crucial role in safeguarding individuals and businesses from financial losses due to medical expenses, but accurately forecasting insurance claims is vital for determining premiums, managing risks, and maintaining profitability. Traditional forecasting methods often fall short in capturing complex patterns in claims behavior. This paper explores the use of machine learning (ML) algorithms, including Ridge Regression, Lasso Regression, Random Forest, and Gradient Boosting, with performance evaluated based on metrics like MSE, RMSE, and R^2 . The analysis uses healthcare insurance data from 2021, identifying key variables influencing claims amounts. Feature selection techniques, such as SHAP and waterfall plots, reveal the most influential factors driving claim amounts, demonstrating how ML can optimize pricing, manage risk, and improve profitability.

Index Terms—Healthcare, insurance, claims, Machine learning

I. INTRODUCTION

Insurance involves transferring the risk of financial loss for a premium, where claims occur when a covered event happens, and the beneficiary is compensated. Forecasting claims is essential for determining adequate premiums, managing financial risk, and ensuring the profitability of insurance companies. Accurate forecasting prevents premiums from being set too low (which would lead to financial loss) or too high (which could result in customer attrition). Insurance pricing traditionally relies on historical claims data, but future incidents determine whether the premiums are sufficient to cover claims. Without accurate forecasting, insurers risk underpricing or overpricing their services. Furthermore, future claims experience plays a key role in projecting long-term profitability, impacting the company's business plans and solvency capital.

Traditional forecasting methods typically involve repeating historical claims patterns or matching them to known probability distributions, where smaller claims occur frequently, and large claims are less common but severe. To improve accuracy, claims are often separated into small and large

categories, with distinct probability distributions for each. ML provides a more advanced alternative for claims forecasting. The application of ML in the insurance sector has grown, particularly in predicting claims amounts and improving operational efficiencies. As noted in [1], ML has the potential to enhance client engagement, fraud detection, asset analysis, and overall scalability in insurance. For example, a hybrid predictive model using Apache Spark and deep learning-based techniques for healthcare readmissions achieved 90.69% accuracy, significantly outperforming traditional models [2].

ML algorithms like Support Vector Machines (SVM), Decision Trees, Random Forests, and Boosting are increasingly employed to forecast average claims amounts and identify the most significant variables in insurance portfolios. Additionally, ML techniques have been successfully applied to detect health insurance fraud in India. The Multi-Layer Feedforward Network model, based on historical claims data, reduces error and detection time, providing significant cost savings [3]. Moreover, the Automated Doctor Machine in rural India uses built-in sensors to diagnose conditions and upload reports to a cloud database, improving healthcare management [4].

Building on these findings, the current study aims to apply ML to real-time insurance data to predict claims amounts, offering a predictive tool for future insurance forecasting.

II. LITERATURE SURVEY

The following section discusses prominent previous literature on insurance data analysis.

This paper [5] compared XGBoost, Gradient Boosting, and Random Forest models for medical insurance premium prediction, highlighting XGBoost's superior performance with the highest R^2 . SHAP and ICE plots were also used to identify key factors influencing premium calculations. Another study [6] introduced an emerging health insurance tool that leveraged big data to streamline operations, automate manual

TABLE I
DESCRIPTION OF INSURANCE CLAIM ATTRIBUTES

S.No	Attribute Name	Description
1	Incurred_Date	The critical date used to determine health insurance benefits.
2	CIm_Date	The date when a healthcare service/treatment was provided to the subscriber/customer.
3	CIm_Settled_Date	The date when the insurance company processes and finalizes your claim, determining the amount they will pay for the services received.
4	Amount_Claimed	The total cost of healthcare services received, billed to the subscriber's insurance company, including all medical expenses related to treatment.
5	Ancillary_Claimed_Amount	Additional benefits beyond standard health insurance coverage, including services like dental and vision care, submitted as part of the claim.
6	Ancillary_Paid_Amount	The portion of the ancillary claimed amount that the insurance company actually pays for specific ancillary services received.
7	Amount_Pending	Any outstanding balance after the insurance company has processed the claim; the difference between the total claimed amount and the amount paid.
8	Amount_Adjusted	Reductions or adjustments made to the original claimed amount due to negotiated rates or contractual agreements.
9	Copay_Amount	A flat fee paid by the subscriber for each medical service, with the insurance company covering the rest.
10	Co_Ins_Amount	A cost-sharing method requiring the member to pay a stated percentage of eligible medical expenses after the deductible has been met.
11	Deductible	The annual amount that must be paid before the insurance policy begins to cover medical care expenses.
12	Status_Change_Date	The date when the status of the healthcare service/treatment changed.
13	Service_Start_Date	The date when the healthcare service/treatment started.
14	Service_End_Date	The date when the healthcare service/treatment ended.
15	Last_Updated_Date	The date of the last updated status of the healthcare service/treatment.

tasks, and enhance policyholder experience, with an ANN-based regression model predicting health insurance premiums with 92.72% accuracy. This research [7] discussed the role of predictive analytics in disease control, emphasizing technological, ethical, and cultural challenges, while advocating for continued collaboration to address global health issues. A study [8] evaluated Stochastic Gradient Boosting and Voting Regressor models in healthcare expense prediction, finding the latter to be the most accurate at 88.98%, considering complexity, interpretability, and efficiency. This work [9] explored data analytics' potential in improving clinical support and operational efficiency in healthcare, while another paper [10] highlighted the transformative impact of data analytics on patient care, operational processes, and the ethical concerns of patient privacy.

This research [11] addressed ethical issues in medical image analysis and AI in healthcare, proposing a framework for ethical data collection, while another study [12] showcased China's digital healthcare transition through telemedicine and Internet Plus Healthcare. A review [13] examined 180 studies on Big Data Analytics in healthcare, emphasizing its potential to enhance diagnostic accuracy and patient outcomes. This paper [14] discussed the challenges of integrating big data analytics in healthcare, particularly around data protection and patient care. Another study [15] developed a regression decision tree algorithm to predict hospitalization days, outper-

forming traditional methods, especially for elderly patients. Research [16] evaluated the impact of supplementary health insurance, while this work [17] focused on Big Data's potential to improve healthcare financing in India. A study [18] examined employee health insurance in Pune SMEs, suggesting tailored policies for employee retention. Research [19] proposed enhancing Montenegro's health insurance system with SMART card integration, and another work [20] introduced data mining techniques to detect fraudulent claims. Further studies explored advancements in health insurance models, including machine learning applications [22], [23], blockchain and IoT for claim accuracy in [24], and predictive models like SARIMA for cost prediction in healthcare (see [25]).

III. METHODOLOGY

This section details the implementation of the Health Insurance Analysis framework, as shown in Figure 1. It outlines the key components, including data collection, preprocessing, Feature selection, model development, and interpretation of results. Data is gathered from diverse sources, such as claims, policyholder demographics, and financial transactions, and integrated into a centralized system. Preprocessing steps like data cleaning and feature engineering ensure the quality of the data. Finally, machine learning models are applied to analyze the data and generate actionable insights for improved insurance decision-making.

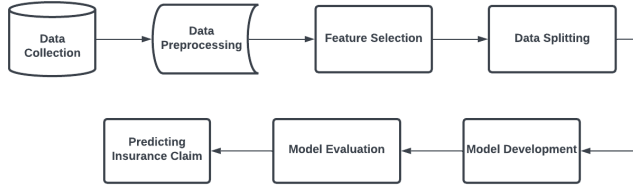


Fig. 1. Architecture Diagram

A. Data Preprocessing

Data preprocessing was conducted to eliminate redundant or irrelevant information that would not contribute to the analysis, ensuring the dataset was streamlined for efficient modeling. During this phase, missing values, outliers, and inconsistent entries were systematically identified and addressed. Since the dataset contained primarily numeric instances, an imputation approach was employed to replace missing values, using techniques such as mean, and median, depending on the data characteristics. Outliers were detected using statistical methods like the Z-score or IQR (Interquartile Range) and were either removed or transformed to prevent skewing the analysis. Inconsistent entries, such as incorrectly formatted data or contradictory values, were corrected based on predefined rules or domain knowledge. Additionally, categorical variables were encoded appropriately to ensure compatibility with machine learning models. These preprocessing steps were essential for ensuring the quality and integrity of the data used in further analysis.

B. Descriptive Statistics

The distributions of the ID variable were carefully estimated using a range of statistical metrics to understand its underlying characteristics and behavior within the dataset. Key metrics included the mean, which provided the central tendency of the values, and the standard deviation and variance, which offered insights into the spread or variability of the ID values. Skewness was computed to assess the asymmetry of the distribution, indicating whether the data leaned more to the left or right of the mean. Kurtosis was analyzed to determine the "tailedness" of the distribution, revealing whether the ID values were more concentrated around the mean or more evenly spread out. Additionally, the cardinality of the ID variable was examined to measure the uniqueness of values, helping to identify whether the variable contained a high number of distinct identifiers or if many repeated entries were present. By evaluating these metrics, a more comprehensive understanding of the ID variable's distribution was obtained, informing subsequent steps in the analysis and model development.

C. Feature Selection Approach

Furthermore, feature selection techniques were systematically employed on the insurance dataset to identify the most relevant attributes that significantly influence claim amounts

and other key outcomes. This step is essential to ensure that only meaningful variables are included in the model, improving predictive accuracy while preventing overfitting and bias. Highly correlated features can lead to multicollinearity, distorting model performance and estimates, which is why this step was crucial in refining the dataset.

The first step in feature selection involved correlation analysis, where pairs of features were examined to identify highly correlated attributes. Features that exhibited a strong linear relationship, typically with a correlation coefficient above a specified threshold, were eliminated to reduce redundancy and potential bias. After removing these highly correlated features, the remaining variables were subjected to more advanced feature selection techniques to assess their importance.

To further refine the feature set, techniques such as Recursive Feature Elimination, Random Forest, and Lasso Regression were employed. RFE is an iterative process that recursively removes the least important features based on model performance, ensuring that only the most relevant attributes are retained. The Random Forest method, a powerful ensemble learning technique, was used to assess the importance of each feature by measuring how much each one contributes to reducing the model's overall prediction error. Additionally, Lasso Regression, which applies L1 regularization, was utilized to shrink the coefficients of less important variables to zero, effectively performing both feature selection and regularization.

D. Regression Analysis

For the features identified during the feature selection phase, regression analysis was conducted to investigate the relationships between the target variable, Insurance Amount Paid, and other relevant features within the dataset. The goal was to build a model that could predict the amount paid for insurance claims based on various factors, such as policyholder demographics, claim history, and other significant attributes. To achieve this, a range of machine learning models were employed, each selected for its strengths in regression tasks. These models included Gradient Boosting, Ridge Regression, Lasso Regression, Decision Tree Regression, Random Forest, Support Vector Regression, XGBoost, K-Nearest Neighbors, and AdaBoost.

To evaluate the performance of these models, two key metrics were used: Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). These metrics provided a quantitative measure of the model's predictive accuracy. Mean Squared Error (MSE), which is defined as:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

where y_i represents the actual values, $\hat{y}_i$ represents the predicted values, and n is the total number of samples. MSE measures the average squared difference between the actual and predicted values, where larger errors are penalized more heavily due to the squaring of the differences. The RMSE, which is the square root of the MSE, is defined as:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

This metric provides a more interpretable error measurement, as it is in the same units as the target variable. RMSE gives an indication of the typical magnitude of the prediction error, with larger errors being more heavily penalized due to the squaring of differences.

Additionally, the R-squared (R^2) value was calculated to assess the proportion of variance in the target variable that is explained by the model. R^2 is given by the formula:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

where $\bar{y}$ is the mean of the actual values. The R^2 score indicates how well the model fits the data, with values closer to 1 indicating a better fit. These performance metrics allowed for a comprehensive evaluation of each model's ability to predict the insurance amount paid, guiding the selection of the best deployment model.

To ensure the evaluation of these models was robust and less susceptible to overfitting, 10-fold cross-validation was employed. In this process, the dataset was divided into 10 equal-sized subsets, or folds. For each fold, one subset was used as the validation set while the model was trained on the remaining 9 subsets. This process was repeated 10 times, with each fold being used as the validation set once. The performance of each model was evaluated by calculating the MAE, RMSE, and R^2 scores for each fold, and the average of these metrics across all folds was computed. This cross-validation approach provided a more reliable assessment of model performance and reduced the likelihood of model evaluation being influenced by a single, potentially unrepresentative partition of the data.

By comparing the results of the models using the MAE, RMSE, and R^2 metrics across the 10 folds, it was possible to assess which algorithm provided the most accurate predictions. This process not only offered insights into the predictive accuracy of each model but also helped identify the relationships between the selected features and the target variable. By refining the model and ensuring it performed well across all folds, we ensured that it was robust enough to make reliable predictions in real-world insurance scenarios. Cross-validation played a critical role in selecting the optimal model for deployment, ensuring that the model would generalize well to new, unseen data.

IV. RESULT

The descriptive statistics in Table II summarize key variables in the dataset. The Amount_Claimed variable has a mean of 1021.82 and a large variance of 79,405.92, indicating a broad distribution with some potential outliers. The variable's skewness of 0.626 suggests a rightward skew, meaning most of the values are clustered toward the lower end of the

range, but there are some higher outliers. Similarly, Ancillary_Claimed_Amount has a mean of 201.63, with higher skewness, further indicating the presence of higher values that skew the distribution to the right. Ancillary_Paid_Amount has a mean of 199.21 and lower skewness, implying a relatively more balanced distribution compared to Ancillary_Claimed_Amount. The Amount_Pending variable has a

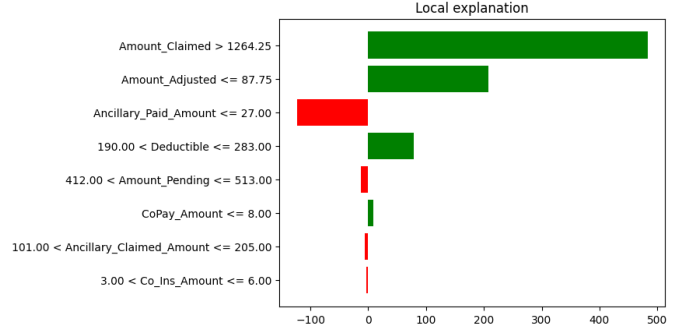


Fig. 2. Lime analysis

mean of 414.35, which, along with a higher standard deviation, shows substantial variation in the amounts still pending for claims. In contrast, Amount_Adjusted shows a mean of 196.83 with lower skewness, indicating a more symmetric distribution. Other variables, such as CoPay_Amount and Deductible, have relatively low means and moderate standard deviations, suggesting less variability in these fields. The Amount_Paid variable has the largest mean and a wide range of values, showing significant variability in the amounts paid for claims.

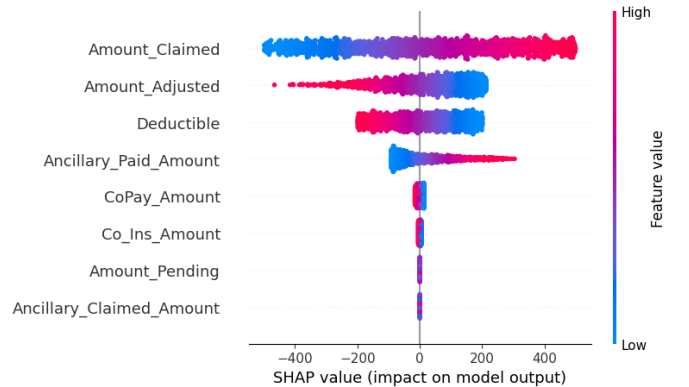


Fig. 3. Shap analysis

Table III compares the performance of several regression models based on Mean MSE, Mean R^2 , and Mean RMSE. Ridge Regression emerges as the most effective model, with a near-perfect R^2 value of 0.92 and an extremely low Mean MSE, indicating that it fits the data exceptionally well with minimal error. Lasso Regression also performs well, demonstrating very high accuracy. However, it slightly lags behind Ridge Regression in terms of Mean MSE. Models like Extra Trees, Decision Tree, and Random Forest follow closely, with slightly higher errors but still show strong

TABLE II
DESCRIPTIVE STATISTICS FOR INSURANCE CLAIMS DATA

Statistic	Mean	Variance	Std	Skewness	Kurtosis	Range
Amount_Claimed	1021.82	79,405.92	281.79	-0.0629	-1.1422	999
Ancillary_Claimed_Amount	201.63	13,374.94	115.65	-0.0105	-1.2049	399
Ancillary_Paid_Amount	101.59	7,992.16	89.40	0.9277	0.0247	395
Amount_Pending	414.35	17,819.08	133.49	0.0594	-0.6532	638
Amount_Adjusted	196.83	18,398.54	135.64	0.6765	-0.1209	690
CoPay_Amount	15.38	74.97	8.66	0.0072	-1.2042	29
Co_Ins_Amount	5.53	8.28	2.88	-0.0133	-1.2157	9
Deductible	290.07	13,154.79	114.69	0.1058	-1.1717	400
Amount_Paid	615.62	108,319.66	329.12	0.2191	-0.7420	1,575

performance, particularly in terms of generalization. Gradient Boosting and Support Vector Regression display respectable performance, but their results are not as strong as those of Ridge and Lasso Regression. XGBoost and K-Nearest Neighbors produce slightly higher RMSE values, with KNN showing the least predictive power among the models tested. AdaBoost, while effective, has the highest Mean MSE and RMSE, suggesting it performs less well compared to the other models.

Fig. 2. presents a line analysis that visualizes the relationship between key features and the model's output. The analysis reveals that Amount_Claimed and Ancillary_Paid_Amount are the most influential features in the prediction model. The graph demonstrates that changes in these features have a significant impact on the model's output, suggesting that they are strong predictors of the target variable. The local segmentation emphasizes how variations in these variables directly affect the regression model's behavior.

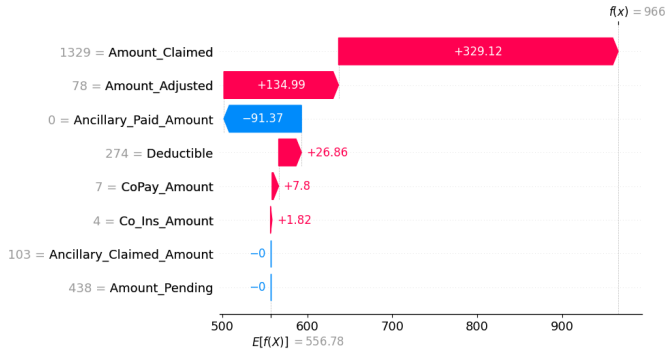


Fig. 4. Waterfall Plot

Fig. 3. illustrates a SHAP (Shapley Additive Explanations) analysis, which provides insight into how each feature contributes to the model's output. The SHAP values indicate that Amount_Claimed and Amount_Adjusted are the most impactful features, driving the model's predictions toward higher values. In contrast, CoPay_Amount and Ancillary_Claimed_Amount exert a negative influence on the predictions, pulling the output lower. This analysis helps

to understand the model's decision-making process and the importance of each feature in determining the final prediction.

TABLE III
MEAN REGRESSION MODEL RESULTS

Model	Mean MSE	Mean R ²	Mean RMSE
Ridge Regression	4.073418e-03	0.920301	60.000064
Lasso Regression	1.367751e-03	0.910000	65.369831
Extra Trees	6.422968e+03	0.893679	80.143420
Decision Tree	4.806760e+03	0.905466	69.330802
Random Forest	1.225022e+03	0.908791	65.000308
Gradient Boosting	8.510110e+02	0.912129	69.172093
Support Vector Regression	1.013831e+04	0.906347	100.689185
XGBoost	9.379614e+02	0.911319	70.626156
K-Nearest Neighbors	1.608164e+03	0.915115	70.101926
AdaBoost	8.793310e+03	0.898398	73.772650

Fig. 4. displays a waterfall plot, which provides a step-by-step breakdown of how each feature contributes to the final prediction. It shows that Amount_Claimed and Amount_Adjusted contribute positively, significantly increasing the model's output, while Deductible has a negative impact, reducing the predicted value. The waterfall plot clearly visualizes how the combination of these features results in the model's final prediction, highlighting the relative influence of each feature.

The results demonstrate that Ridge Regression is the most effective model for predicting insurance claim outcomes, with minimal error and near-perfect accuracy. SHAP and waterfall analyses provide valuable insights into the relative importance of each feature, highlighting Amount_Claimed and Amount_Adjusted as the key drivers of the model's predictions. The line analysis further emphasizes the significant relationship between these features and the target variable. These findings offer a comprehensive understanding of the model's performance and the critical factors influencing insurance claim outcomes.

V. CONCLUSION

Ridge Regression emerged as the top-performing model, achieving a near-perfect R² of 0.92, indicating an excellent fit with minimal error. Lasso Regression also performed highly, but slightly lagged behind Ridge due to a higher Mean MSE. Models like Random Forest and Extra Trees showed strong generalization performance, though with marginally higher

errors. Key predictors in the model included Amount Claimed and Amount Adjusted, which strongly influenced the final predictions, driving them toward higher values. Conversely, features such as CoPay Amount and Ancillary Claimed Amount had a negative impact on the model's predictions, pulling the output lower.

REFERENCES

- [1] Seely, S., 2018. Eight Use Cases for Machine Learning in Insurance. Azure. Available online: <https://azure.microsoft.com/en-us/blog/eight-use-cases-for-machine-learning-in-insurance/> (accessed on 28 December 2022).
- [2] N. J. Ravindran and P. Gopalakrishnan, "Predictive Analysis for Healthcare Sector Using Big data Technology," 2018 Second International Conference on Green Computing and Internet of Things (ICGCIoT), Bangalore, India, 2018, pp. 326-331, doi: 10.1109/ICGCIoT.2018.8753090.
- [3] Sankar, K.H. and Shyam, A.V., Customer Ethics In Health Insurance Claims: A Study Using Genetic Algorithm.
- [4] Aswath, G.I., Vasudevan, S.K. and Sampath, N., 2020. A frugal and innovative telemedicine approach for rural India—automated doctor machine. *International Journal of Medical Engineering and Informatics*, 12(3), pp.278-290.
- [5] Orji, U. and Ukwandu, E., 2024. Machine learning for an explainable cost prediction of medical insurance. *Machine Learning with Applications*, 15, p.100516.
- [6] Kaushik, K., Bhardwaj, A., Dwivedi, A.D. and Singh, R., 2022. Machine learning-based regression framework to predict health insurance premiums. *International journal of environmental research and public health*, 19(13), p.7898.
- [7] Ogugua, J.O., Onwumere, C., Arowoogun, J.O., Anyanwu, E.C., Odilibe, I.P. and Akomolafe, O., 2024. Data science in public health: A review of predictive analytics for disease control in the USA and Africa. *World Journal of Advanced Research and Reviews*, 21(1), pp.2753-2769.
- [8] Kandula, A.R., Kalyanapu, S., Rayapalli, S.N., Veerabathina, K.R., Modugumudi, V. and Kanikella, S.R., 2024, March. Medical Insurance Predictive Modelling: An Analysis of Machine Learning Methods. In 2024 IEEE International Conference on Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI) (Vol. 2, pp. 1-5). IEEE.
- [9] Arowoogun, J.O., Babawarun, O., Chidi, R., Adeniyi, A.O. and Okolo, C.A., 2024. A comprehensive review of data analytics in healthcare management: Leveraging big data for decision-making. *World Journal of Advanced Research and Reviews*, 21(2), pp.1810-1821.
- [10] Ibeh, C.V., Elufioye, O.A., Olorunsogo, T., Asuzu, O.F., Nduubuisi, N.L. and Daraojimba, A.I., 2024. Data analytics in healthcare: A review of patient-centric approaches and healthcare delivery. *World Journal of Advanced Research and Reviews*, 21(2), pp.1750-1760.
- [11] Padmapriya, S.T. and Parthasarathy, S., 2024. Ethical data collection for medical image analysis: A structured approach. *Asian Bioethics Review*, 16(1), pp.95-108.
- [12] Hu, Z., Qin, X., Chen, K., Huang, Y.N., Wang, R.S., Tung, T.H., Chuang, Y.C. and Wang, B.L., 2024. Chinese Health Insurance in the Digital Era: Bibliometric Study. *Interactive Journal of Medical Research*, 13(1), p.e52020.
- [13] Ahmed, A., Xi, R., Hou, M., Shah, S.A. and Hameed, S., 2023. Harnessing big data analytics for healthcare: A comprehensive review of frameworks, implications, applications, and impacts. *IEEE Access*.
- [14] Singh, R.K., Agrawal, S., Sahu, A. and Kazancoglu, Y., 2023. Strategic issues of big data analytics applications for managing health-care sector: a systematic literature review and future research agenda. *The TQM Journal*, 35(1), pp.262-291.
- [15] Y. Xie et al., "Predicting Days in Hospital Using Health Insurance Claims," in *IEEE Journal of Biomedical and Health Informatics*, vol. 19, no. 4, pp. 1224-1233, July 2015, doi: 10.1109/JBHI.2015.2402692.
- [16] H. Yang, "Is Supplementary Health Insurance an Effective Supplement to Basic Health Insurance in China? a Two-part Model and Quantile Approach," 2021 International Conference on Public Management and Intelligent Society (PMIS), Shanghai, China, 2021, pp. 308-311, doi: 10.1109/PMIS52742.2021.00075.
- [17] S. Gupta and P. Tripathi, "An emerging trend of big data analytics with health insurance in India," 2016 International Conference on Innovation and Challenges in Cyber Security (ICICCS-INBUSH), Greater Noida, India, 2016, pp. 64-69, doi: 10.1109/ICICCS.2016.7542360.
- [18] A. Gurbaxani and R. Raman, "Perception of SMEs in India towards Employee Health Insurance Plan," 2022 International Conference on Sustainable Islamic Business and Finance (SIBF), Sakhr, Bahrain, 2022, pp. 205-208, doi: 10.1109/SIBF56821.2022.9939961.
- [19] A. Ivanović and P. Raković, "E-health Card Information System: Case Study Health Insurance Fund of Montenegro," 2019 8th Mediterranean Conference on Embedded Computing (MECO), Budva, Montenegro, 2019, pp. 1-5, doi: 10.1109/MECO.2019.8760284.
- [20] Kareem, S., Ahmad, R.B. and Sarlan, A.B., 2017, November. Framework for the identification of fraudulent health insurance claims using association rule mining. In 2017 IEEE Conference on Big Data and Analytics (ICBDA) (pp. 99-104). IEEE.
- [21] M. R. Islam et al., "MHIVis: Visual Analytics for Exploring Mental Illness of Policyholders in Life Insurance Industry," 2020 7th International Conference on Behavioural and Social Computing (BESC), Bournemouth, United Kingdom, 2020, pp. 1-4, doi: 10.1109/BESC51023.2020.9348301.
- [22] A. Vinora, V. Surya, E. Lloyds, B. Kathir Pandian, R. N. Deborah and A. Gobinath, "An Efficient Health Insurance Prediction System using Machine learning," 2023 International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems (ICES), Chennai, India, 2023, pp. 1-5, doi: 10.1109/ICES60034.2023.10465334.
- [23] P. Singh, K. D. Singh, V. Tripathi and V. Chaudhari, "Use of Ensemble Based Approach to Predict Health Insurance Premium at Early Stage," 2022 International Conference on Computational Intelligence and Sustainable Engineering Solutions (CISES), Greater Noida, India, 2022, pp. 566-569, doi: 10.1109/CISES54857.2022.9844398.
- [24] Inayatulloh, R. Toyib, E. Sahputra, Y. Reswan, Y. Darmi and Meisyarah, "Adoption Block Chain Technology and Internet of Thing for Medical Record in Health Insurance," 2022 13th International Conference on Information and Communication Technology Convergence (ICTC), Jeju Island, Korea, Republic of, 2022, pp. 1539-1542, doi: 10.1109/ICTC55196.2022.9952723.
- [25] K. W. Wedanage, R. Wickramasuriya, K. T. Win and P. Perez, "Forecasting healthcare cost in Australia using health insurance claims data," 2021 17th International Computer Engineering Conference (ICENCO), Cairo, Egypt, 2021, pp. 35-39, doi: 10.1109/ICENCO49852.2021.9698885.
- [26] Y. Xu and M. Shi, "Research on optimal mixed health insurance contact based on risk sharing," 2016 International Conference on Industrial Economics System and Industrial Security Engineering (IEIS), Sydney, NSW, Australia, 2016, pp. 1-5, doi: 10.1109/IEIS.2016.7551880.
- [27] L. Xiaogun and L. Run, "An Improved K-means Clustering Model Based on Support Vector Machine for Health Insurance Cost Prediction," 2022 IEEE 2nd International Conference on Electronic Technology, Communication and Information (ICETCI), Changchun, China, 2022, pp. 521-525, doi: 10.1109/ICETCI55101.2022.9832085.
- [28] H. Pingping, "Analysis of Urban Employee Basic Medical Insurance in Beijing," 2023 16th International Conference Management of large-scale system development (MLSD), Moscow, Russian Federation, 2023, pp. 1-5, doi: 10.1109/MLSD58227.2023.10303931.